Orbital Advection Across Azimuthal Level Jumps in the AthenaK Shearing Box

Validation Document

`athenak-multigrid` tree, branch `fargo-amr`
parent commit `d3edb948` (branch `dust-multigrid`)

October 28, 2026

Abstract

Orbital advection (FARGO) in the AthenaK shearing box now accepts x_2 neighbors one level coarser or finer, so refinement may cover only part of the azimuth (method: `fargo_amr_implementation.pdf`). This document records the tests. Every annular and uniform configuration of the earlier test suite, hydro and MHD, is bit-identical to the parent code. With the orbital-advection flux correction the remap conserves mass to round-off across static and moving level jumps ($\sim 5 \times 10^{-14}$ over hundreds of steps; 10^{-7}–10^{-6} without it). Serial and 3-, 4- and 7-rank runs are bit-identical. A spatially uniform epicycle stays exactly uniform on a refined patch. Vortical shearing waves on a static patch and under a moving box lie between the ring and coarse-grid results, with no visible artifact at the level jumps. Swing amplification with multigrid self-gravity on a patch moves the peak 0.60% toward linear theory, and a gravity-off control shows that gravity adds no deviation of its own. A timestep limit lets a run proceed where the shift would otherwise exceed the neighbor's rows. The production test — SC14 clump-following AMR with and without rings — is still to be run.

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1 Test ladder

All runs are on a Mac (Apple Silicon, serial build and Open MPI build), October 28, 2026. Run directories are in `validation/run/fargo_amr/`; its `README.md` lists the command for each one. "Patch" means a refined region that covers only part of x_2, so it has x_2 level jumps; "ring" means the old annular configuration.

Test	What it probes	Result
Regression (§2)	same-level path unchanged	bit-identical
Epicycle on a patch (§3)	buffers, indices, prolongation	exact
Conservation (§4)	flux correction, static and AMR	5×10^{-14}
MPI (§5)	per-sub-face messages and tags	bit-identical
Vortical shearing wave (§6)	structure through level jumps	between ring and coarse
Swing with gravity (§7)	multigrid at x_2 level jumps	+0.60% toward theory
Timestep limit (§8)	large shifts	runs, conserves
SC14 production (§9)	cost	not yet run

2 Regression: nothing changes where nothing jumps

On a same-level face the new code performs the same arithmetic as before, only with a different buffer layout and depth. Each input below was run with a build of the parent commit and with the new build, and the binary dumps (the data after the parameter header, which differs by the added input line) and history files compared byte by byte. All are **bit-identical**:

Input (`inputs/shearing_box/`)	Configuration
`epicycle_smr_ring`	level-1 ring, 1 orbit
`epicycle_amr_ring`, `epicycle_amr_ring2`	moving ring, 1 and 2 levels, incl. block create/destroy sequence
`hydro_incompress_shwave_smr_ring{,2}`	ring, 1 and 2 levels
`hydro_incompress_shwave_amr_ring`	moving ring
`epicycle`, `hydro_incompress_shwave`, `hydro_compress_shwave`	uniform mesh (VTK dumps for the shwaves)
`mhd_compress_shwave`	uniform mesh, MHD with orbital advection

The AMR ring inputs were run with `oa_annular = true`, which they need for the ring-wise flag synchronization (now off by default). The new orbital-advection timestep limit does not bind in any of them.

3 The epicycle on a patch

`epicycle_smr_ypatch.athinput`: $20 \times 20 \times 1$ box, $128^2 \times 4$ root cells, $32^2 \times 4$ blocks; the level-1 region covers $|x|, |y| < 5$, two of the four block rows in y, so four x_2 faces are level jumps. The `ipert=1` epicycle is spatially uniform at all times, so any error in buffer layout, indexing or prolongation appears as $O(1)$ structure. After one orbit (365 steps):

- density, all velocity components and the internal energy are uniform to the last digit of the single-precision output in every one of the 44 blocks;
- every history column matches the ring run (`epicycle_smr_ring`), and the mass does not change;
- on 4 MPI ranks every dump is bit-identical to the serial run.

4 Exact conservation

A uniform field cannot show a conservation error, and the vortical shwave of §6 has amplitude 10^{-4} and constant density. The conservation test is therefore the compressible shwave (`hydro_compress_shwave_smr`, isothermal, HLLE, PLM) at amplitude 0.1 on a 4×4 box, $128^2 \times 4$ root cells, $32^2 \times 4$ blocks, with a level-1 patch over $|x|, |y| < 1$. The x_1 boundaries are plain *periodic*: the shear-periodic remap at the x_1 faces is itself conservative only to $\sim 10^{-8}$ and would hide the effect. Only the hydro update and the orbital remap then move mass. The moving-box variant uses the AMR criterion of the implementation document with $x_c = 0.9 \sin t$, $y_c = 0.8 \sin 2t + t$ and half-widths 0.3.

maximum $\lvert M(t) - M(0)\rvert / M(0)$	flux correction on	off
static patch, $t = 0$–2 (453 steps)	5.2×10^{-14}	1.8×10^{-7}
moving box (AMR), $t = 0$–4 (899 steps; 98 blocks created, 84 destroyed)	5.7×10^{-14}	1.0×10^{-6}
static patch, shear-periodic x_1	3.8×10^{-9}	1.1×10^{-7}

With the correction the remap is conservative to round-off, through regridding too (Fig. 1); the total y-momentum stays at 3×10^{-16} against 5×10^{-7} without. With shear-periodic x_1 boundaries the correction still removes most of the drift; the remaining 4×10^{-9} is the boundary remap.

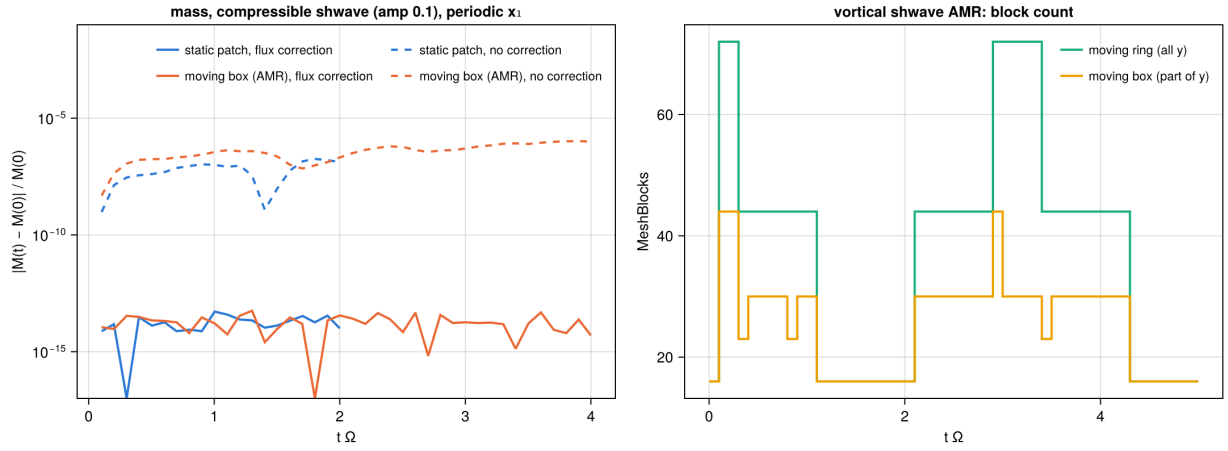


Figure 1: Left: relative mass drift of the compressible shwave (amplitude 0.1, periodic x_1) on a static patch (blue) and under a moving box with AMR (orange), with (solid) and without (dashed) the orbital-advection flux correction. Right: MeshBlock count of the vortical-shwave AMR runs of §6 — the moving box refines part of the ring's blocks. Script `figures/plot_fargo_amr_tests.jl`.

5 MPI

The conservation runs (static patch and moving box) were repeated on 3, 4 and 7 ranks. All binary dumps are bit-identical to the serial runs. The history files differ only in the volume sums, by the MPI reduction order: $\leq 6.5 \times 10^{-13}$ on a mass of 16, and $\sim 10^{-15}$ in momenta whose value is itself round-off. Mass is conserved to 1×10^{-14} on every rank count. The vortical-shwave moving box on 4 ranks is likewise bit-identical to the serial run in all 51 dumps.

6 Vortical shearing wave through static and moving level jumps

The Johnson & Gammie (2005) vortical shwave (`ipert=2`, isothermal, PPM; $64^2 \times 4$ root cells on 0.5×0.5, $16^2 \times 4$ blocks, $t = 0$–5) on five meshes, compared with a uniform 128^2 run as the reference:

run	refined region (level 1)	peak x-KE vs 128^2	max $\lvert\Delta \mathrm{KE}_x\rvert$/peak
uniform 64^2	—	-4.5%	1.1×10^{-1}
ring, static	$\lvert x\rvert < 0.125$, all y	-7.9%	8.2×10^{-2}
patch, static	$\lvert x\rvert, \lvert y\rvert < 0.125$	-8.1%	9.3×10^{-2}
moving ring, AMR	$x_c = 0.2 \sin t$	-12.2%	1.3×10^{-1}
moving box, AMR	$+\, y_c = 0.15 \sin t + 0.1t$	-11.9%	1.3×10^{-1}

Inputs: `hydro_incompress_shwave{,_smr_ring,_smr_ypatch,_amr_ring,_amr_box}`; all history output at full precision (`output1/data_format=%.15e`). The patch lies between the ring and the coarse grid in the history error, and the moving box matches the moving ring (the box creates and destroys 77 blocks, the ring 112; Fig. 1, right). The peak error of the ring is larger than that of the coarse grid; this is a property of the ring runs, not of this work.

Figure 2 maps the v_x error against the 128^2 run. Inside the static patch the error is larger than inside the ring at $t = 1$: the wave reaches the fine region after crossing coarse cells in y, while in the ring it never leaves the fine grid. At the x_2 level jumps the error bands run straight across, with no stripes or kinks. Averaged over the refined x_1 columns of the patch, the error next to the jumps lies between those of the fine interior and the coarse rows ($t = 1$: 9.1, 4.9 and 11×10^{-2} of max $\lvert v_x\rvert$; at $t = 2.5$ the three agree to 10%).

The planned twin without orbital advection is not available for patches: with `orbital_advection = false`, the patch mesh collapses at the first step ($\Delta t \to 10^{-45}$) *in the parent code as well*, with or without `prolong_primitives`. The ring mesh runs. This is a separate, pre-existing problem (§9).

7 Swing amplification with multigrid gravity on a patch

`tests/swing_selfgrav_smr_ypatch.athinput` is the swing test of `multigrid_sb_validation.pdf` (Phase 2a: $2\pi \times 2\pi \times \pi/4$, $128^2 \times 16$ root cells, 16^3 blocks, $4\pi G\rho_0 = 1.9$, a leading wave with $k_x/k_y = -6$) with the level-1 region cut to the middle four of eight block rows in y (six of eight columns in x, as for the ring). The x_2 faces at $y = \pm\pi/2$ are then level jumps for orbital advection and for the multigrid octets. The ring and uniform runs were repeated with the new code.

run	peak $d_{\cos}/A$	peak time	max $\lvert d_{\sin}\rvert/A$
uniform, multigrid	-3.511418	7.804	7.3×10^{-14}
ring	-3.583481	7.804	1.3×10^{-13}
patch	-3.532422	7.804	6.1×10^{-13}

The uniform and ring runs reproduce the Phase-2a values to the last digit. The patch moves the peak 0.60% toward the converged answer (linear theory 3.605), under a third of the ring's 2.05%, with half the ring's refined area; the phase lock stays at machine level. The gravity-off control (`hydro_srcterms/self_gravity=false`) attributes the shift as in Phase 2a: the patch-vs-uniform deviation is larger without gravity (5.8×10^{-2} maximum, peak +1.11%) than with it (3.6×10^{-2}, +0.60%), so multigrid gravity across the x_2 level jumps adds no deviation of its own (Fig. 3).

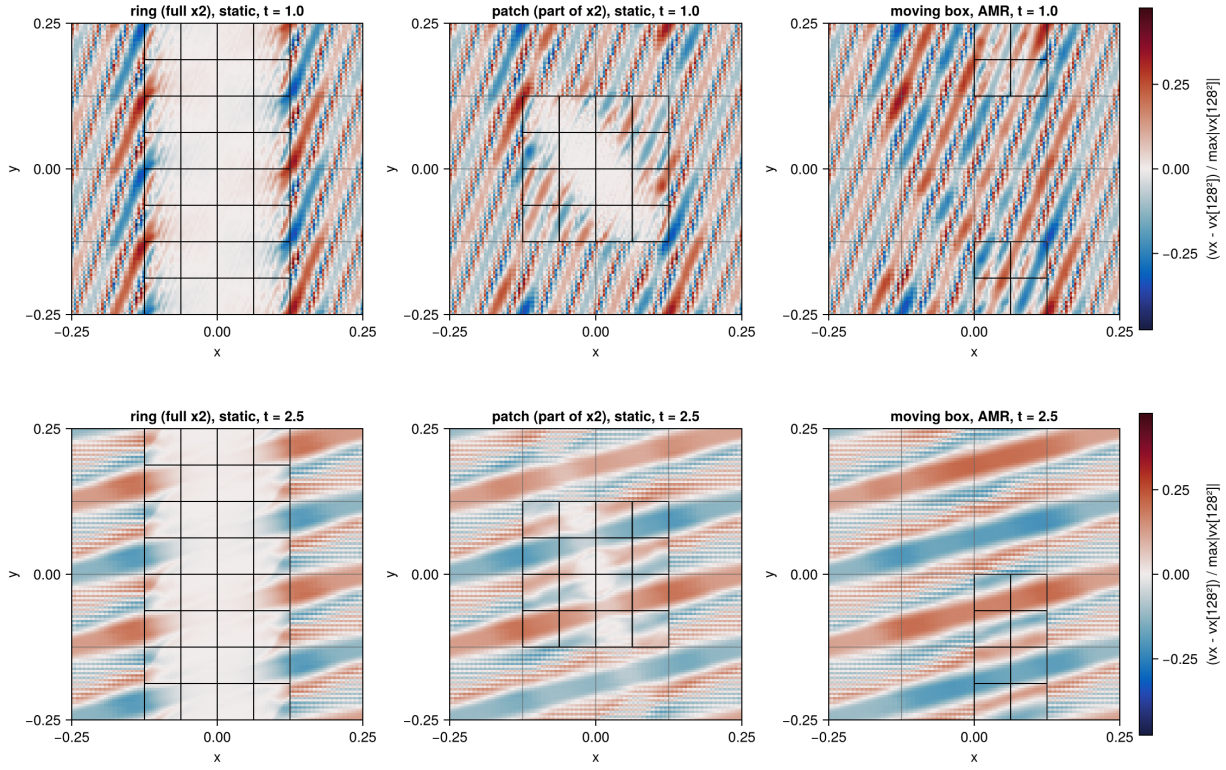


Figure 2: v_x error of the vortical shwave against a uniform 128^2 run, normalized to $\max|v_x|$, averaged over z, shown at the finest resolution (coarse cells repeated), at $t = 1$ (top) and $t = 2.5$ (bottom). Left: ring. Middle: static patch, x_2 level jumps at $y = \pm 0.125$. Right: moving box (AMR). Black: refined blocks; grey: root blocks. The checkerboard in coarse regions is the coarse cells against the fine reference. Script `figures/plot_fargo_amr_shwave.jl`.

8 The timestep limit

A cold variant of the static-patch conservation test (sound speed 0.05, amplitude 10^{-3}) lets the hydro timestep grow until the shift at $|x| = 2$ would be 8 cells against 6 available rows. The build without the limit stops at the first step with the guard's message. With it, Δt is held at $0.999\,(6-2)\,\Delta x_2/(q\Omega \max|x|) = 0.0416$, exactly the predicted value; the run reaches $t = 8$ (193 steps) and conserves mass to 8×10^{-14}.

9 Not yet validated

- **SC14 production.** The clump-following AMR of the gravito-turbulent box (`gravito_turb_l64z16_clump_amr_overlay.athinput`) restarted from an archived $\beta = 3$ state, once with `oa_annular = true` and once without: block count, cost per cycle and whether the timestep limit binds. Needs a cluster.
- **GPU.** The kernels follow the existing team-scratch pattern but have run only on the CPU (serial and MPI).

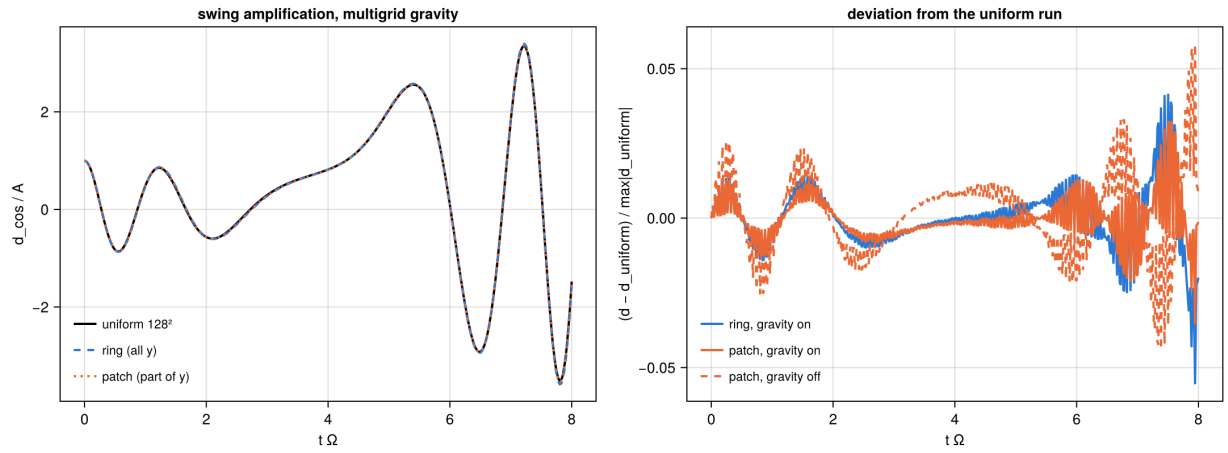


Figure 3: Swing amplification with multigrid self-gravity. Left: projection of the density onto the shearing wave, $d_{\cos}/A$, for the uniform, ring and patch meshes (indistinguishable at this scale). Right: deviation from the uniform run, with gravity (solid) and, for the patch, without (dashed). The fast oscillation of the deviations is present in the ring run as well. Script `figures/plot_fargo_amr_tests.jl`.

- **MHD.** Orbital advection with refinement remains fatal for MHD.
- **Non-FARGO refinement with x_2 level jumps** fails in the parent code (§6); not addressed here.

10 Reproduction

```
cd validation/run/fargo_amr        # README.md: one command per run directory
python3 compare.py unif128 unif ring ypatch amrring amrbox   # shwave table
python3 errmap.py 00010                                      # error at the level jumps
(cd swing && python3 cmp.py)                                 # swing table
cd ../.. && julia figures/plot_fargo_amr_shwave.jl           # Fig. 2
julia figures/plot_fargo_amr_tests.jl                        # Figs. 1 and 3
```